# Practice workbook · RAM, ROM, virtual memory and flash memory

The function and properties of RAM, ROM and virtual memory, why a computer needs both RAM and ROM, what is stored in ROM, and how flash differs from RAM.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T4-05

# 1. Question 2 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

**2** Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

**(1)**

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

# 1. Question 2 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

**(2)**

(ii) Describe how data is read from a magnetic hard drive.

**(3)**

OLevels.COM SCHOOL

(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

| Stage | Sequence number |
|---|---|
| The instruction is placed onto the data bus | |
| The Arithmetic and Logic Unit (ALU) performs a calculation | |
| The address of the next instruction is placed onto the address bus | |
| The instruction is decoded by the control unit | |

**(4)**

## 1. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 2. Question 1 · part d · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

**1** Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

**(3)**



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

**(1)**

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

**(2)**

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

## 2. Question 1 · part d · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

**(Total for Question 1 = 11 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 2. Question 1 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 3. Question 3 · part b.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

**3** Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

**(2)**

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

**(2)**

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

**(2)**

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

**(2)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 4. Question 5 · part d.iii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.iii only.

**5** Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

**(2)**

(ii) Give **one** reason why a patient may not want to use this online service.

**(1)**

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

**(1)**

(ii) Name the type of network used to access the server from the Stockton building.

**(1)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

# 4. Question 5 · part d.iii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

OLevels.com SCHOOL

(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

| Description | RAM | ROM |
|---|---|---|
| Stores the boot up sequence | | |
| The contents are lost when the laptop is shut down | | |

**(Total for Question 5 = 12 marks)**

## 4. Question 5 · part d.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

**5** Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

**(2)**

| Requirement | Example |
|---|---|
| Input | |
| Process | Calculate the path the bee will move along to its new position |
| Output | |

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

## 5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

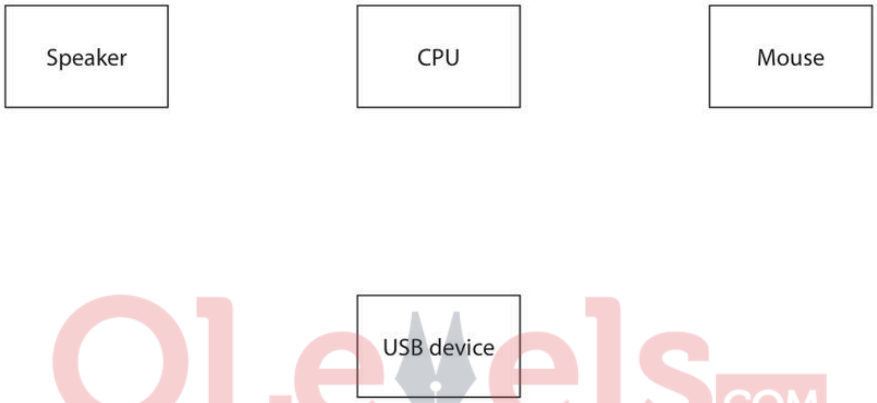
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

**(3)**




OLevels.COM SCHOOL

# 5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

## 5. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# 6. Question 2 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

**2** Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

**(2)**

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

**(1)**

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

OLevels.COM — SCHOOL —

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

**(2)**

.....

.....

.....

.....

## 6. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# 7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

**4** A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

**(2)**

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

**(2)**

.....

.....

.....

.....

(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

**(2)**

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

## 7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

**(2)**

| | 8-bit binary pattern | Denary value |
|---|---|---|
| Lowest value | | |
| Highest value | | |

OLevels.com SCHOOL

# 7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

| O | S | P | O AND S | NOT P | (O AND S) AND (NOT P) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

OLevels.com SCHOOL

(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

.....

.....

.....

.....

**(Total for Question 4 = 13 marks)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WR

## 7. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL